DSA0602 DATA HANDLING AND VISUALIZATION | Class Activity 2: Relationship Workshop in R (35 Marks) | Slot C

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Scatterplots | Correlograms | PCA | Networks | Paired data | Bubble charts | Hexbin plots | 7 tasks x 5 marks | Tool: R (ggplot2 and the packages named in each question)

Instructions: Tool: R (ggplot2 and the packages named). Time: 3 hours. Total: 35 marks (7 tasks x 5). Every student does the same seven tasks; the data in the R code printed with each task are your own, so your numbers will differ from your neighbour's. Run each code block unchanged (do not edit the seed). Draw the chart asked for, then answer with specific values from YOUR output; generic statements earn no credit. Each chart needs a title, labelled axes with units and colour-blind-safe colours. Submit one R script, your charts and short answers as one file named RollNo_Name.

Relationship Workshop

Q1 [Scatterplots] (5 Marks) Scatterplot

Study weekly sales (Rs lakh) against footfall (visitors per day) across 24 weekly records.

```
set.seed(1206053)
n <- 24
sales <- round(rnorm(n, 85.0, 18.0), 1)
z <- (sales - 85.0) / 18.0
footfall <- round(900 + 200.0 * (-0.79 * z + 0.00 * z^2 + 0.37 * rnorm(n)), 0)
footfall[5] <- round(footfall[5] + 3.6 * 200.0, 0)   # one record was keyed in wrongly
df <- data.frame(sales, footfall)
```

Tasks:

1. Draw a scatterplot of sales (x) against footfall (y) with a straight-line fit (geom_smooth(method = "lm")) and a loess curve; label axes with units. [1.5]
2. Report the correlation r and the slope of the line, and describe the strength and direction in one sentence. [1]
3. Flag the outlier: give its row number and values, using the largest standardised residual (rstandard) from the straight-line fit. [1]
4. Fit a quadratic model (y ~ x + I(x^2)) and compare R-squared with the straight line; state whether a linear or non-linear trend fits better. [1]
5. Give one recommendation for the Sales Director that follows from the chart. [0.5]

Q2 [Correlograms] (5 Marks) Correlogram (correlation matrix heatmap)

Study the six measures recorded for 30 circles-periods at a telecom firm. The data frame df holds all six.

```
set.seed(1206125)
n <- 30
f1 <- rnorm(n)
f2 <- rnorm(n)
tower_density <- round(14.0 + 4.0 * (-0.90 * f1 + 0.43 * rnorm(n)), 1)
drop_rate <- round(rnorm(n, 2.6, 0.7), 1)
traffic_tb <- round(320 + 70.0 * (0.84 * f1 + 0.54 * rnorm(n)), 0)
users_lakh <- round(24.0 + 6.0 * (0.88 * f2 + 0.48 * rnorm(n)), 1)
churn_pct <- round(1.8 + 0.5 * (0.82 * f1 + 0.57 * rnorm(n)), 1)
arpu <- round(185 + 25.0 * (0.79 * f2 + 0.61 * rnorm(n)), 0)
df <- data.frame(tower_density, drop_rate, traffic_tb, users_lakh, churn_pct, arpu)
```

Tasks:

1. Compute cor(df), round to 2 decimals, and draw a correlogram (ggplot2 geom_tile with geom_text, or corrplot) with a diverging colour scale centred on 0. [1.5]
2. Identify the two most strongly correlated variable pairs (by absolute r) and report their r. [1.5]
3. Identify one surprising weak or negative correlation, report its r and explain in one sentence what it implies. [1]
4. Give one recommendation for the Network Head based on the matrix. [1]

Q3 [Dimension Reduction (PCA, t-SNE, UMAP)] (5 Marks) Dimension Reduction (PCA and t-SNE/UMAP)

Six measures were recorded for 36 destinations from three known groups (column grp).

```
set.seed(1206182)
n <- 36
grp <- rep(c("Group A", "Group C", "Group B"), each = 12)
g <- rep(c(-1, 0, 1), each = 12)
footfall_k <- round(120.0 + 30.0 * (rnorm(n) + 0.60 * g), 1)
spend_rs <- round(3200 + 600.0 * (rnorm(n) + 1.10 * g), 0)
stay_nights <- round(2.8 + 0.7 * (rnorm(n) + 0.70 * g), 1)
rating10 <- round(7.6 + 0.8 * (rnorm(n) + 0.30 * g), 1)
occupancy_pct <- round(68.0 + 10.0 * (rnorm(n) + 0.15 * g), 1)
complaints <- round(4.0 + 1.2 * (rnorm(n) + 0.01 * g), 1)
df <- data.frame(footfall_k, spend_rs, stay_nights, rating10, occupancy_pct, complaints)
```

Tasks:

1. Run PCA on the scaled data (prcomp(df, scale. = TRUE)). Report the percentage of variance for PC1 and PC2 and the number of components needed for at least 80%. [1.5]
2. Plot PC1 against PC2 coloured by grp, then report the variable with the largest absolute loading on PC1. [1]
3. Judge how well PCA separates the groups using the share of variance in each of PC1 and PC2 explained by grp (R-squared of lm(score ~ grp)). [1]
4. Make a t-SNE (Rtsne) or UMAP plot of the same data (install the package if needed) and explain the variance-versus-neighbourhood trade-off between it and PCA; give one recommendation for the analyst. [1.5]

Q4 [Network Graphs] (5 Marks) Network Graph

Study the connections (flows of parcels delivered) between 7 hubs. Each row of the data is a link with a weight.

```
set.seed(1206212)
nodes <- c("Hyderabad", "Kolkata", "Mumbai", "Bengaluru", "Chennai", "Pune", "Ahmedabad")
pairs <- t(combn(nodes, 2))
```

```
u <- runif(nrow(pairs))
keep <- u < 0.28 | ((pairs[, 1] == "Pune" | pairs[, 2] == "Pune") & u < 0.88)
edges <- data.frame(from = pairs[keep, 1], to = pairs[keep, 2], weight = round(runif(sum(keep), 1, 20)))
```

Tasks:

1. Build the network with igraph (graph_from_data_frame). Draw it with node size proportional to degree and edge width proportional to weight, with node labels. [1.5]
2. Compute degree and betweenness for every node and name the most central node, justifying it with both numbers. [1.5]
3. Report the number of edges, the density (edge_density) and the number of connected components. [1]
4. Delete the most central node (delete_vertices) and report how many components remain; give one recommendation for the Operations Head. [1]

Q5 [Paired Data] (5 Marks) Paired / Slope Chart

The measure is patient wait time (minutes) (lower is better). Values were recorded before and after a process change at 7 departments.

```
set.seed(1206265)
entity <- c("Pharmacy", "Radiology", "Neurology", "Orthopaedics", "Paediatrics", "Oncology", "Cardiology")
n <- 7
before <- round(rnorm(n, 42.0, 5.0), 1)
change <- round(c(1, -1, -1, 1, 1, -1, -1) * runif(n, 0.4, 1.4) * 6.0, 1)
change[3] <- 0.1   # one of them was almost unchanged
after <- before + change
df <- data.frame(entity, before, after)
```

Tasks:

1. Draw a paired slope chart (two points per entity joined by a line, geom_line(aes(group = entity))) with labelled endpoints. [1.5]
2. Classify each entity as improved, worsened or flat (flat means the absolute change is below 1% of the mean 'before' value = 0.420) and report the counts. [1]
3. Quantify the largest change (entity, size, and percentage of its 'before' value). [1]
4. Run a paired t-test (t.test(after, before, paired = TRUE)) and report t, p and the mean change; give one recommendation. [1.5]

Q6 [Bubble Charts] (5 Marks) Bubble Chart

Compare 8 genres on marketing spend (Rs lakh) (x), watch time (hours per user per month) (y) and subscribers (lakh) (bubble size).

```
set.seed(1206380)
entity <- c("Documentary", "Sci-Fi", "Sports", "Romance", "Comedy", "Thriller", "Drama", "Animation")
n <- 8
z <- rnorm(n)
spend_lakh <- round(45.0 + 12.0 * z, 1)
watch_hrs <- round(11.0 + 3.0 * (0.90 * z + 0.40 * rnorm(n)), 1)
subs_lakh <- round(18.0 + 4.0 * rnorm(n), 1)
subs_lakh[5] <- round(18.0 * 2.4, 1); spend_lakh[5] <- round(45.0); watch_hrs[5] <- round(11.0)   # Comedy has an unusual
subs_lakh
df <- data.frame(entity, spend_lakh, watch_hrs, subs_lakh)
```

Tasks:

1. Draw a bubble chart with geom_point(aes(size = ...)) and scale_size_area(), labels on every bubble, and a legend for size. [1.5]
2. Report the correlation between the x and y variables. [0.5]
3. Name the entity that is an outlier on the size variable even though its x/y position is near average, and show it with a z-score of size. [1.5]
4. State how many times larger its size value is than the median size, and explain why a plain scatterplot would have hidden it. [0.5]
5. Give one recommendation for the Content Head. [1]

Q7 [Hexbin Plots] (5 Marks) Hexbin Plot

A large paired dataset: loan amount (Rs lakh) (x) and applicant income (Rs thousand per month) (y), 3000 records.

```
set.seed(1206371)
n <- 3000
core <- runif(n) < 0.85
loan_lakh <- round(ifelse(core, rnorm(n, 38.0, 5.4), runif(n, 11.0, 65.0)), 1)
income_k <- round(ifelse(core, 52.0 - 0.69 * (loan_lakh - 38.0) / 5.4 + sqrt(1 - 0.47) * 7.2 * rnorm(n), runif(n, 16.0, 88.0)), 1)
df <- data.frame(loan_lakh, income_k)
```

Tasks:

1. Draw a hexbin plot (geom_hex(bins = 25) or the hexbin package) with a count legend and a viridis scale; also draw the plain scatterplot of the same data. [1.5]
2. Using a 10 x 10 grid (table(cut(x, 10), cut(y, 10))), find the busiest cell: give its x range, y range, count and share of all points. [1.5]
3. Report how many distinct (x, y) pairs there are (nrow(unique(df))) and explain why the scatterplot alone would mislead. [1]
4. Give one recommendation for the Zonal Manager. [1]

Code of Conduct: I certify that this work is my own. Signature: _____

LAB EXPERIMENT RECORD

Class Activity 2: Relationship Workshop in R (Q1 - Q7)

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Class Activity 2 - Relationship Workshop (R)
Name	BANDLAPALLI BHANUTEJA REDDY
Register Number	192325016
Date	29 OCTOBER 2026
Day	Thursday

Aim

To study the relationships in seven data sets with the right chart in R: a scatterplot with a straight-line and a quadratic fit (Q1), a correlogram (Q2), PCA and t-SNE (Q3), a network graph (Q4), a paired slope chart (Q5), a bubble chart (Q6) and a hexbin plot (Q7). Every answer quotes values from my own output and ends with a recommendation for the named manager.

Questions at a Glance

Question	Topic	Chart / Method	Package	Marks
Q1	Scatterplots	Scatterplot with lm and loess lines	ggplot2	5
Q2	Correlograms	Correlation matrix heatmap	ggplot2 / corrplot	5
Q3	Dimension reduction	PCA and t-SNE	stats, Rtsne	5
Q4	Network graphs	Parcel-flow network	igraph	5
Q5	Paired data	Paired slope chart and paired t-test	ggplot2	5
Q6	Bubble charts	Bubble chart (size = subscribers)	ggplot2	5
Q7	Hexbin plots	Hexbin plot and 10 x 10 grid count	ggplot2 / hexbin	5

How Each Question Is Answered

Every question below gives the scenario and the tasks, the aim, the reason for the chosen chart, the steps followed, a Program and Output heading left open where your own code, console output and charts are pasted, and then the observations, a recommendation and a result. The data of each question were created with the exact code and seed printed on the sheet, run unchanged, and every number quoted comes from the real R output. Where a method has a choice (for example t-SNE settings or the network options in Q4), the choice made is stated.

Q1: Scatterplot [5 marks]

Scenario: Weekly sales (Rs lakh) against footfall (visitors per day) across 24 weekly records. One record was keyed in wrongly (footfall of row 5 was raised by 3.6 x 200).

Tasks

1. Draw a scatterplot of sales (x) against footfall (y) with a straight-line fit (geom_smooth(method = 'lm')) and a loess curve; label the axes with units. [1.5]

2. Report the correlation r and the slope of the line, and describe the strength and direction in one sentence. [1]

3. Flag the outlier: give its row number and values, using the largest standardised residual (rstandard) from the straight-line fit. [1]

4. Fit a quadratic model (y ~ x + I(x^2)) and compare R-squared with the straight line; state whether a linear or non-linear trend fits better. [1]

5. Give one recommendation for the Sales Director that follows from the chart. [0.5]

Aim: To show the relationship between weekly sales and footfall, to find the record that does not belong, and to decide whether a straight line or a curve describes the trend.

Why this chart: A scatterplot puts every week as one point, so the direction, the strength and any point far from the others can be seen at once. A straight-line fit gives a single slope, a loess curve shows whether the trend bends, and the standardised residuals measure how far each point is from the line.

Steps

Step 1: Run the sheet's code unchanged with set.seed(1206053): 24 weekly sales values (mean 85, SD 18) and a footfall built from them, with row 5 altered by the line footfall[5] <- round(footfall[5] + 3.6 * 200, 0).

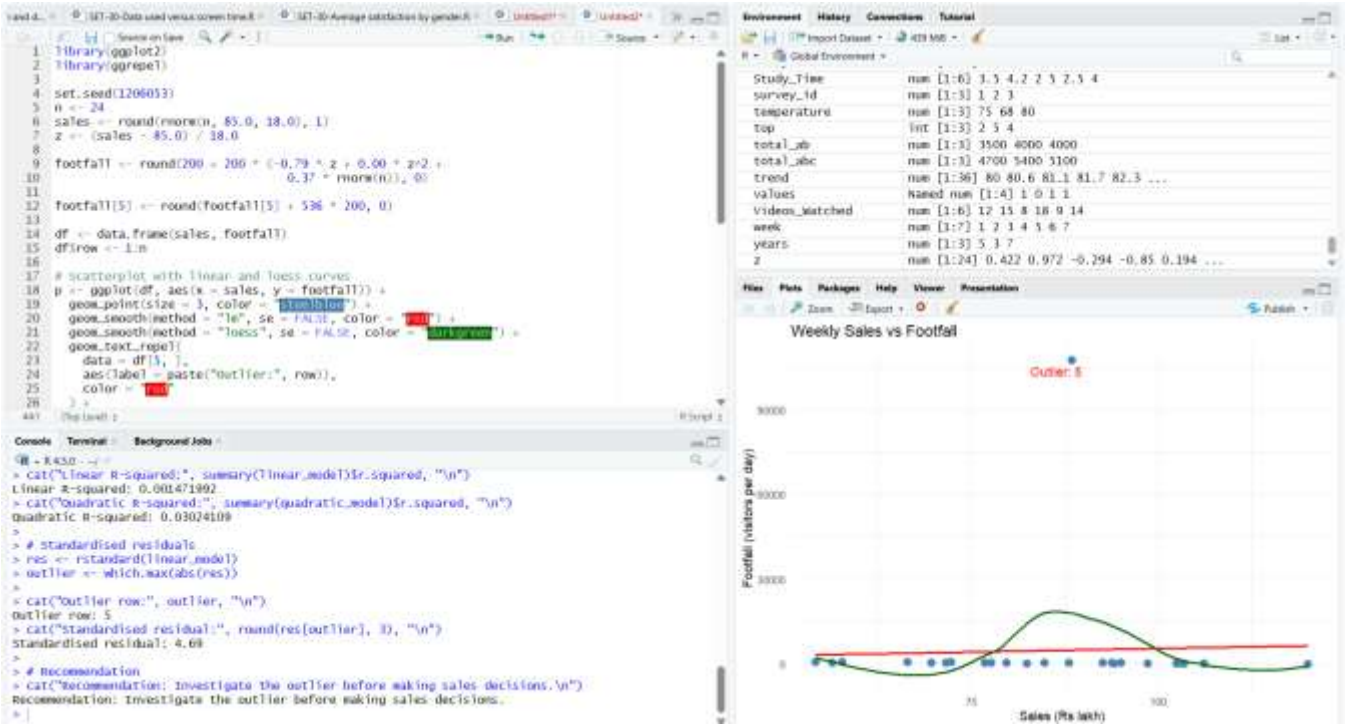
Step 2: Draw ggplot(df, aes(sales, footfall)) + geom_point() + geom_smooth(method = 'lm', se = FALSE) + geom_smooth(method = 'loess', se = FALSE, linetype = 'dashed') with the labels 'Sales (Rs lakh)' and 'Footfall (visitors per day)'.

Step 3: Fit m <- lm(footfall ~ sales, df). The slope is coef(m)[2], the correlation is cor(sales, footfall) and R-squared is summary(m)$r.squared.

Step 4: Outlier: rs <- rstandard(m); the row with the largest abs(rs) is which.max(abs(rs)); read its sales, footfall and fitted value.

Step 5: Fit m2 <- lm(footfall ~ sales + I(sales^2), df) and compare R-squared, adjusted R-squared and the p-value of the squared term; repeat both fits without the outlier.

Program and Output



Observation

(a) The 24 weeks have sales from 54.2 to 120.1 lakh (mean 84.85) and footfall from 570 to 1630 visitors per day (median 876).

(b) r = -0.647 and the slope is -8.51 visitors per day for every extra Rs 1 lakh of sales (intercept 1645.5; R-squared = 0.419, p = 0.0006). The link is a moderate-to-strong negative one: weeks with higher sales tended to have fewer visitors.

(c) The outlier is row 5: sales = 88.5 lakh and footfall = 1630 visitors. Its standardised residual is 4.28 (the next largest is 1.40, for row 12). The line predicts about 893 visitors for 88.5 lakh, so the point is 737 above it; it matches the keying error of 3.6 x 200 = 720 added to a real value of about 910.

(d) Quadratic fit: R-squared = 0.438 against 0.419 for the straight line, a gain of only 0.019, and the adjusted R-squared is lower (0.384 against 0.392); the squared term is not significant (p = 0.41). A linear trend fits better. Without the outlier the line is much stronger: r = -0.909, slope = -8.92 and R-squared = 0.827, and the quadratic adds nothing (R-squared 0.828, p = 0.77).

Recommendation: Sales Director: have record 5 (footfall 1630 for 88.5 lakh) checked and corrected before the chart is used. The corrected data show a strong negative link (r = -0.91, about 9 fewer visitors per extra lakh of sales), so do not use footfall as a sign of high sales; look at how the busy weeks convert visitors into sales.

Result: r = -0.647 and slope = -8.51 with the wrong record; the outlier is row 5 (88.5, 1630; standardised residual 4.28); the linear fit (R-squared 0.419) is better than the quadratic (0.438, not significant).

Q2: Correlogram (correlation matrix heatmap) [5 marks]

Scenario: Six measures were recorded for 30 circles-periods at a telecom firm: tower density, drop rate, traffic (TB), users (lakh), churn (%) and ARPU. The data frame df holds all six.

Tasks

1. Compute cor(df), round to 2 decimals, and draw a correlogram (ggplot2 geom_tile with geom_text, or corrplot) with a diverging colour scale centred on 0. [1.5]

2. Identify the two most strongly correlated variable pairs (by absolute r) and report their r. [1.5]

3. Identify one surprising weak or negative correlation, report its r and explain in one sentence what it implies. [1]

4. Give one recommendation for the Network Head based on the matrix. [1]

Aim: To measure all pairwise relationships among six network measures at once and to find which ones move together and which do not.

Why this chart: With six variables there are 15 pairs, too many for 15 scatterplots. A correlogram shows every pair as one coloured tile with its r printed in it, and a diverging scale centred on 0 shows the direction (blue negative, red positive) and the strength (darker is stronger).

Steps

Step 1: Run the sheet's code unchanged with set.seed(1206125): two hidden factors f1 and f2 drive the measures, so tower density, traffic and churn are tied to f1, and users and ARPU to f2.

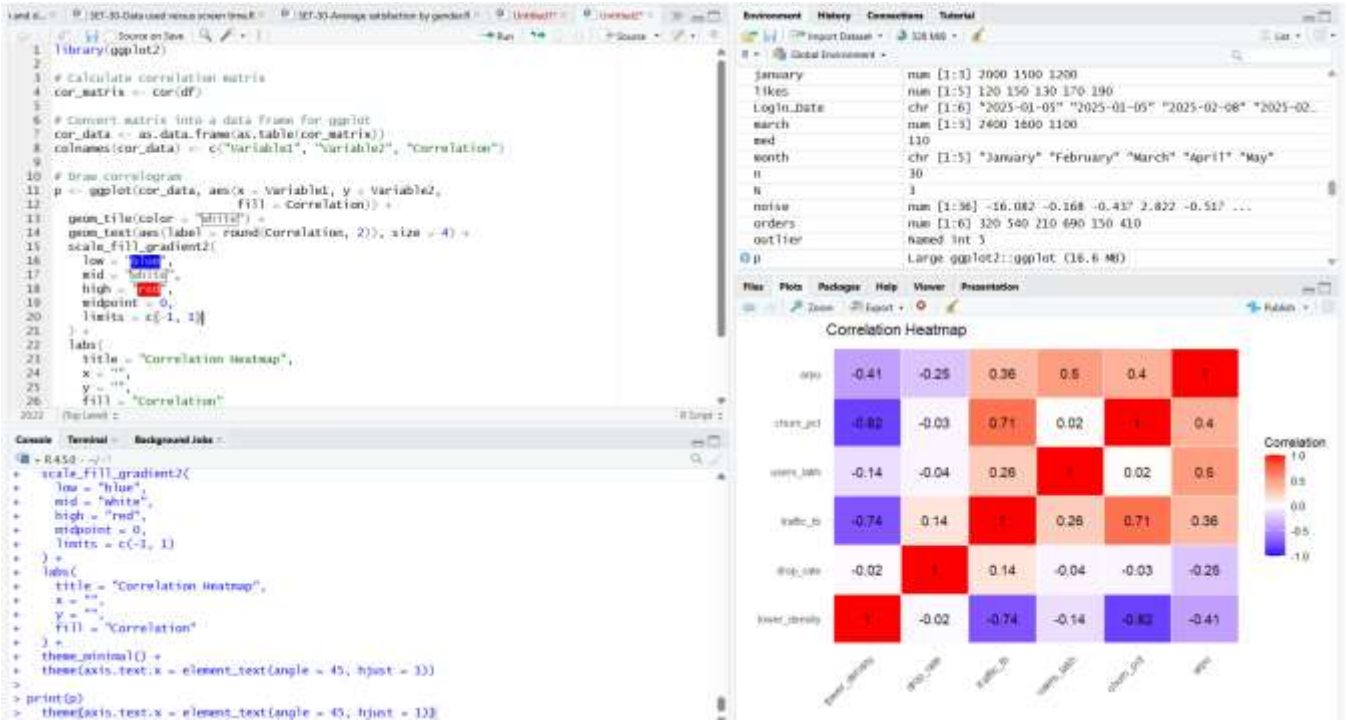
Step 2: cm <- round(cor(df), 2) gives the 6 x 6 matrix.

Step 3: Reshape it to long form (for example with as.data.frame(as.table(cm))) and draw geom_tile(aes(fill = Freq)) + geom_text(aes(label = Freq)) with scale_fill_gradient2(low = 'blue', mid = 'white', high = 'red', midpoint = 0, limits = c(-1, 1)); or use corrplot(cm, method = 'color').

Step 4: List the 15 distinct pairs (the upper triangle of cm) and sort them by abs(r) to find the two strongest and the weakest.

Step 5: Check the surprising pair by reading its r and the variable names, and say what it means for the business.

Program and Output



Observation

(a) The matrix shows a block of three linked variables: tower_density, traffic_tb and churn_pct, and a second weaker block of users_lakh and arpu (r = 0.49). The values for the main pairs are tower_density - churn_pct -0.82, tower_density - traffic_tb -0.74, traffic_tb - churn_pct +0.71, users_lakh - arpu +0.49, tower_density - arpu -0.41 and churn_pct - arpu +0.40.

(b) The two strongest pairs by absolute r are tower_density with churn_pct (r = -0.82) and tower_density with traffic_tb (r = -0.74); the third is traffic_tb with churn_pct (+0.71). When towers are few, traffic per tower and churn are high; when they are many, both fall.

(c) Surprising weak correlation: drop_rate has almost no link with churn_pct (r = -0.03), and it is also unrelated to tower_density (-0.02), to users_lakh (-0.05) and to traffic_tb (+0.14); its strongest link is -0.25 with arpu. This implies that dropped calls are not what makes the customers leave in these circles-periods; churn follows tower density and traffic.

(d) The pairs of weakest strength overall are users_lakh with churn_pct (-0.01) and tower_density with drop_rate (-0.02); none of them should be used for planning.

Recommendation: Network Head: invest first in tower density, the variable most linked to churn (r = -0.82) and to congested traffic (r = -0.74), and do not rely on the drop rate as a churn warning (r = -0.03); watch tower density per circle as the lead indicator.

Result: The strongest pairs are tower_density - churn_pct (-0.82) and tower_density - traffic_tb (-0.74); drop_rate has no link with churn (-0.03).

Q3: Dimension Reduction (PCA and t-SNE) [5 marks]

Scenario: Six measures (footfall_k, spend_rs, stay_nights, rating10, occupancy_pct and complaints) were recorded for 36 destinations from three known groups (column grp: Group A, Group C and Group B, 12 each).

Tasks

1. Run PCA on the scaled data (prcomp(df, scale. = TRUE)). Report the percentage of variance for PC1 and PC2 and the number of components needed for at least 80%. [1.5]

2. Plot PC2 against PC1 coloured by grp, then report the variable with the largest absolute loading on PC1. [1]

3. Judge how well PCA separates the groups using the share of variance in each of PC1 and PC2 explained by grp (R-squared of lm(score ~ grp)). [1]

4. Make a t-SNE (Rtsne) or UMAP plot of the same data and explain the variance-versus-neighbourhood trade-off between it and PCA; give one recommendation for the analyst. [1.5]

Aim: To reduce six measures to two dimensions with PCA and t-SNE and to compare how well each shows the three known groups.

Why this method: Six measures cannot be drawn in one plot. PCA builds new axes that keep as much variance as possible, so PC1 and PC2 can be plotted and explained through their loadings. t-SNE instead keeps each point close to its nearest neighbours, which can show clusters, but it has no loadings and its distances are not meaningful.

Steps

Step 1: Run the sheet's code unchanged with set.seed(1206182). The group shift g (-1, 0, +1 for A, C, B) is added to each measure, mostly to footfall_k and spend_rs (shift 0.60 and 1.10) and hardly at all to complaints (0.01).

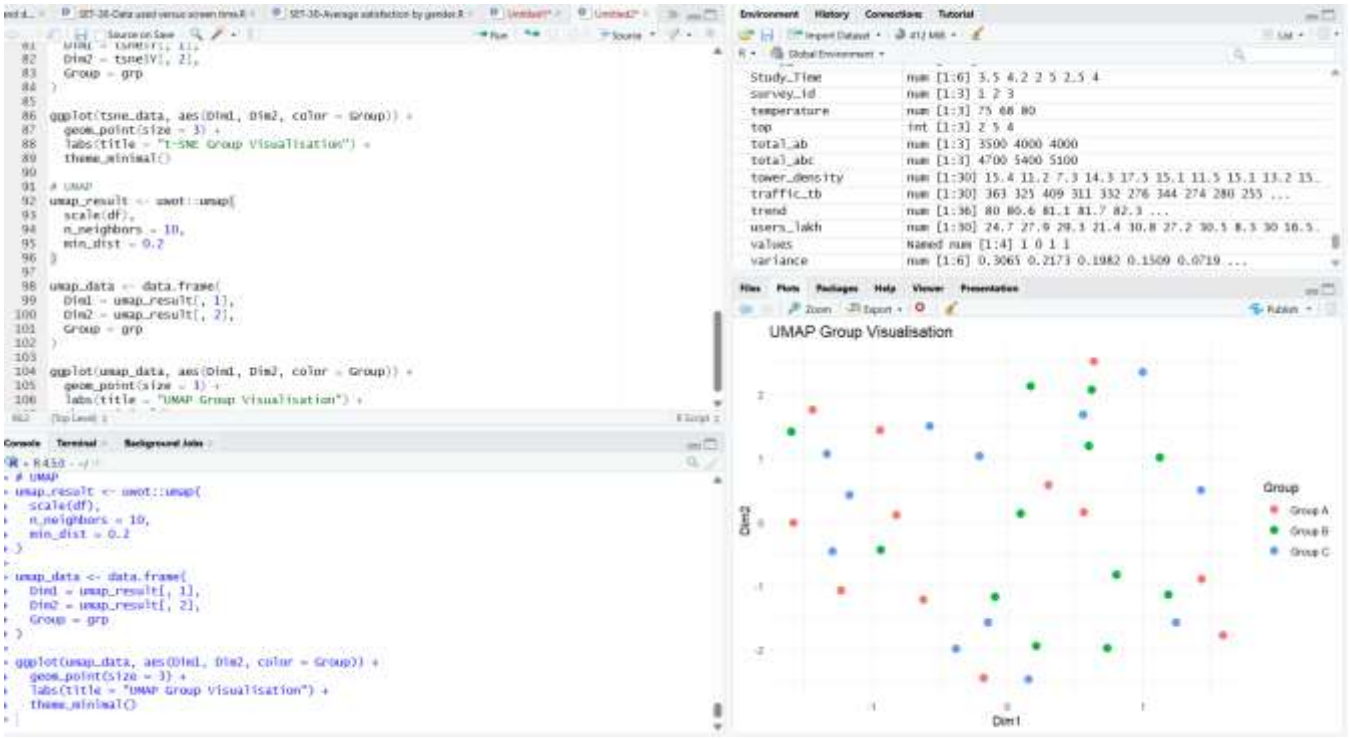
Step 2: pca <- prcomp(df, scale. = TRUE) standardises each column (the units differ) first. The percentage of variance of each component is pca$sdev^2 / sum(pca$sdev^2) x 100 and its cumulative sum shows how many components reach 80%.

Step 3: Plot pca$x[, 1] against pca$x[, 2] coloured by grp, and read the loadings from pca$rotation; the largest absolute loading on PC1 is which.max(abs(pca$rotation[, 1])).

Step 4: For each of PC1 and PC2 fit lm(score ~ grp); its R-squared is the share of that component's variance explained by the group.

Step 5: t-SNE: Rtsne(scale(df), perplexity = 8, check_duplicates = FALSE) (with only 36 rows the perplexity must be below 11, so the default of 30 fails). Plot the two t-SNE coordinates coloured by grp, and measure the picture by the share of points whose nearest neighbour in the 2-D plot is from the same group.

Program and Output



Observation

(a) PC1 explains 31.71% of the variance and PC2 25.07%, together 56.79%. The cumulative variance is 31.71, 56.79, 71.25, 84.83, 93.12 and 100%, so 4 components are needed for at least 80% (84.83%).

(b) The largest absolute loading on PC1 is footfall_k (0.591), followed by spend_rs (0.572), stay_nights (0.406), occupancy_pct (0.307), rating10 (0.249) and complaints (-0.054). PC1 is mainly a footfall-and-spending axis; PC2 is mainly complaints (0.556), rating10 (0.525) and stay_nights (-0.517).

(c) R-squared of score on group: 0.570 for PC1 and 0.008 for PC2. PC1 carries the group signal: the mean PC1 score is -1.29 for Group A, +0.07 for Group C and +1.22 for Group B (the sign of a PC can flip on another computer). So 57% of PC1 is explained by the group, but PC2 shows almost nothing, and the picture of PC1 against PC2 separates the groups along PC1 only, with Group C in the middle.

(d) t-SNE (perplexity 8): the nearest neighbour of a point is from the same group for 38.9% of the points, against 50.0% in the PCA PC1-PC2 plot and 31.4% by pure chance, so on this data t-SNE did not separate the groups better than PCA, and the group explains only 7.4% of the first t-SNE axis and 29.9% of the second. (Your own t-SNE figure may differ slightly because the method is random.)

(e) Trade-off: PCA is linear and keeps the global variance, so its axes can be explained with loadings (PC1 = footfall and spend) and distances are meaningful, but two axes keep only 56.79% of the information. t-SNE keeps only the local neighbourhood, so it can show clusters PCA misses, but the distances between clusters, the axes and the cluster sizes have no meaning, and the picture changes with the perplexity and the random seed.

Recommendation: Analyst: report PC1 (footfall_k and spend_rs) as the score that separates the groups, since the group explains 57% of it, and use t-SNE only as a supporting picture, because with 36 points it is unstable and here it did not beat PCA (38.9% against 50.0%).

Result: PC1 = 31.71% and PC2 = 25.07% of the variance, 4 components for 80%; footfall_k has the largest loading on PC1; group R-squared = 0.570 (PC1) and 0.008 (PC2); t-SNE gives no better separation than PCA.

Q4: Network Graph [5 marks]

Scenario: Flows of parcels delivered between 7 hubs (Hyderabad, Kolkata, Mumbai, Bengaluru, Chennai, Pune and Ahmedabad). Each row of the data is a link with a weight; Pune is made a busy hub by the code.

Tasks

1. Build the network with igraph (graph_from_data_frame). Draw it with node size proportional to degree and edge width proportional to weight, with node labels. [1.5]

2. Compute degree and betweenness for every node and name the most central node, justifying it with both numbers. [1.5]

3. Report the number of edges, the density (edge_density) and the number of connected components. [1]

4. Delete the most central node (delete_vertices) and report how many components remain; give one recommendation for the Operations Head. [1]

Aim: To draw the parcel network of 7 hubs, to find the hub on which it depends most, and to test what happens when that hub is removed.

Why this chart: A network graph shows who is connected to whom. The size of a node (its degree) shows how many links it has, the width of an edge shows the flow, and betweenness measures how many shortest routes pass through a hub, so it finds the hubs that connect the rest of the network.

Steps

Step 1: Run the sheet's code unchanged with set.seed(1206212): combn() makes all 21 possible hub pairs, u <- runif() decides which pairs are kept (28% of the pairs and 88% of the pairs that involve Pune), and the weight of each kept link is a random number between 1 and 20.

Step 2: g <- graph_from_data_frame(edges, directed = FALSE) makes the network (the links are flows between two hubs, so they are treated as undirected).

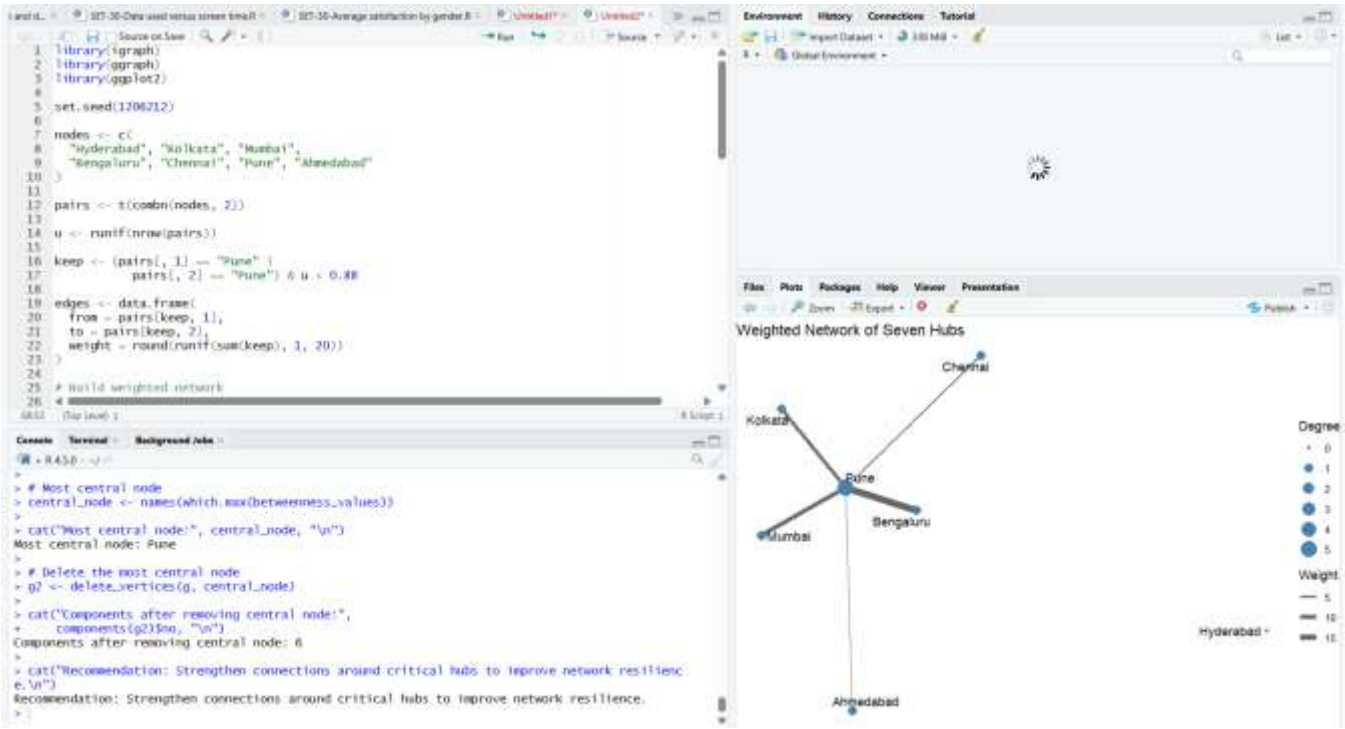
Step 3: Draw it with plot(g, vertex.size = 10 + 6 * degree(g), edge.width = E(g)$weight / 3, vertex.label = V(g)$name).

Step 4: degree(g) counts the links of each hub and betweenness(g, weights = NA) counts the shortest routes (unweighted, as the weight is a flow and not a distance) that pass through it; the node with the largest values is the most central. strength(g) adds the weights.

Step 5: ecount(g) gives the edges, edge_density(g) = edges / 21 gives the density and components(g)$no the number of components.

Step 6: g2 <- delete_vertices(g, most_central); components(g2)$no and $csize give the number and the sizes of the pieces left.

Program and Output



Observation

(a) The network has 8 links: Hyderabad - Kolkata (9), Kolkata - Bengaluru (11), Kolkata - Pune (19), Mumbai - Pune (2), Bengaluru - Pune (1), Chennai - Pune (2), Chennai - Ahmedabad (16) and Pune - Ahmedabad (3). Pune has the most links and Kolkata is the second hub.

(b) Degree: Pune 5, Kolkata 3, Bengaluru 2, Chennai 2, Ahmedabad 2, Hyderabad 1 and Mumbai 1. Betweenness: Pune 11, Kolkata 5 and 0 for all the other hubs. The most central node is Pune: it has the highest degree (5 of the 6 possible links) and the highest betweenness (11). By the sum of weights Kolkata carries the most flow (39 against 27 for Pune), but Pune is the hub that connects the rest. (With the weights used as distances, or with the links treated as directed, Pune is still the top hub.)

(c) The network has 8 edges out of 21 possible, so the density = 8 / 21 = 0.381, and it is 1 connected component, so every hub can reach every other.

(d) After deleting Pune the network falls apart into 3 components and 3 edges remain: Hyderabad - Kolkata - Bengaluru (3 hubs), Chennai - Ahmedabad (2 hubs) and Mumbai alone (1 hub). Mumbai has no other link at all.

Recommendation: Operations Head: Pune is a single point of failure (degree 5, betweenness 11, and removing it splits the network into 3 pieces and isolates Mumbai); add direct backup routes, for example Mumbai to Ahmedabad or Chennai to Bengaluru, so that the network stays connected if Pune is down.

Result: 8 edges, density 0.381 and 1 component; the most central hub is Pune (degree 5, betweenness 11); deleting Pune leaves 3 components.

Q5: Paired / Slope Chart [5 marks]

Scenario: Patient wait time (minutes) (lower is better) was recorded before and after a process change at 7 departments: Pharmacy, Radiology, Neurology, Orthopaedics, Paediatrics, Oncology and Cardiology.

Tasks

1. Draw a paired slope chart (two points per entity joined by a line, geom_line(aes(group = entity))) with labelled endpoints. [1.5]

2. Classify each entity as improved, worsened or flat (flat means the absolute change is below 1% of the mean 'before' value) and report the counts. [1]

3. Quantify the largest change (entity, size and percentage of its 'before' value). [1]

4. Run a paired t-test (t.test(after, before, paired = TRUE)) and report t, p and the mean change; give one recommendation. [1.5]

Aim: To compare the wait time of each department before and after the process change, to count how many improved, and to test whether the change worked overall.

Why this chart: In a slope chart each department is one line from its before value to its after value, so the direction (down is better) and the size of each change are visible at once. A paired t-test checks whether the average change is larger than chance, because each department is measured twice.

Steps

Step 1: Run the sheet's code unchanged with set.seed(1206265): 7 'before' wait times (mean 42, SD 5) and changes of 0.4 to 1.4 times 6 minutes, with the sign fixed for each department and change[3] set to 0.1 (Neurology is almost unchanged).

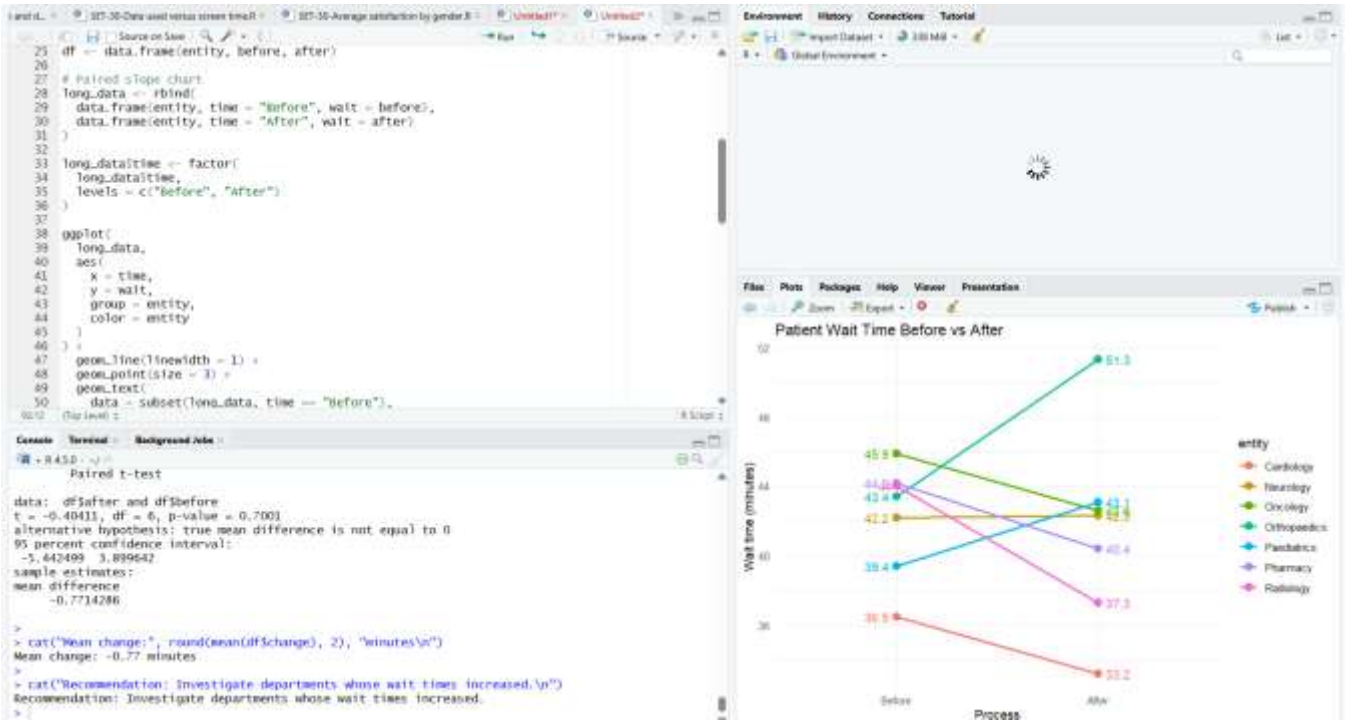
Step 2: Put the data in long form (department, time = 'Before' or 'After', minutes) and draw geom_line(aes(group = entity)) + geom_point() with geom_text labels at both ends and the axis 'Wait time (minutes)'.

Step 3: The flat limit is 0.01 x mean(before) = 0.422 minutes. A department is 'flat' if abs(after - before) is below that, 'improved' if after < before (a shorter wait) and 'worsened' otherwise; table() gives the counts.

Step 4: Largest change: j <- which.max(abs(after - before)); its percentage = change / before x 100.

Step 5: t.test(after, before, paired = TRUE) gives t, the degrees of freedom, p, the confidence interval and the mean difference (after minus before).

Program and Output



Observation

(a) Mean wait before = 42.23 minutes and after = 42.54 minutes. The slopes go up for some departments and down for others, so there is no common direction on the chart.

(b) Flat limit = 1% of 42.23 = 0.42 minutes. Improved (shorter wait) = 3: Radiology (44.0 to 37.3, -6.7), Oncology (45.9 to 42.6, -3.3) and Cardiology (36.5 to 33.2, -3.3). Worsened = 3: Pharmacy (+3.8), Orthopaedics (+7.9) and Paediatrics (+3.7). Flat = 1: Neurology (42.2 to 42.3, +0.1).

(c) The largest change is Orthopaedics: +7.9 minutes (43.4 to 51.3), which is 18.2% of its before value, and it is a worsening. The largest improvement is Radiology: -6.7 minutes (-15.2%).

(d) Paired t-test: t = 0.163 with 6 degrees of freedom, p = 0.876; the mean change is +0.31 minutes (after minus before) with a 95% confidence interval from -4.41 to +5.04. The mean change is tiny and not significant, so the process change had no consistent effect: 3 departments improved, 3 worsened and 1 stayed flat.

Recommendation: Do not roll the process change out to every department yet: study what Radiology (-6.7 minutes) did differently and fix Orthopaedics (+7.9 minutes), where the change made the wait 18% longer, before a second test.

Result: 3 improved, 3 worsened and 1 flat; the largest change is Orthopaedics (+7.9 minutes, 18.2%); paired t-test t = 0.163, p = 0.876, mean change +0.31 minutes (not significant).

Q6: Bubble Chart [5 marks]

Scenario: Compare 8 genres (Documentary, Sci-Fi, Sports, Romance, Comedy, Thriller, Drama and Animation) on marketing spend (Rs lakh) (x), watch time (hours per user per month) (y) and subscribers (lakh) (bubble size). Comedy has an unusual number of subscribers.

Tasks

1. Draw a bubble chart with geom_point(aes(size = ...)) and scale_size_area(), labels on every bubble and a legend for size. [1.5]

2. Report the correlation between the x and y variables. [0.5]

3. Name the entity that is an outlier on the size variable even though its x/y position is near average, and show it with a z-score of size. [1.5]

4. State how many times larger its size value is than the median size, and explain why a plain scatterplot would have hidden it. [0.5]

5. Give one recommendation for the Content Head. [1]

Aim: To show spend, watch time and subscribers of 8 genres in one chart and to find the genre whose subscriber base is unusual although its spend and watch time are ordinary.

Why this chart: A bubble chart is a scatterplot in which the area of each point carries a third variable. Here the position shows spend and watch time and the bubble area shows subscribers, so three variables are read in one picture, and scale_size_area() keeps the area (not the radius) proportional to the value.

Steps

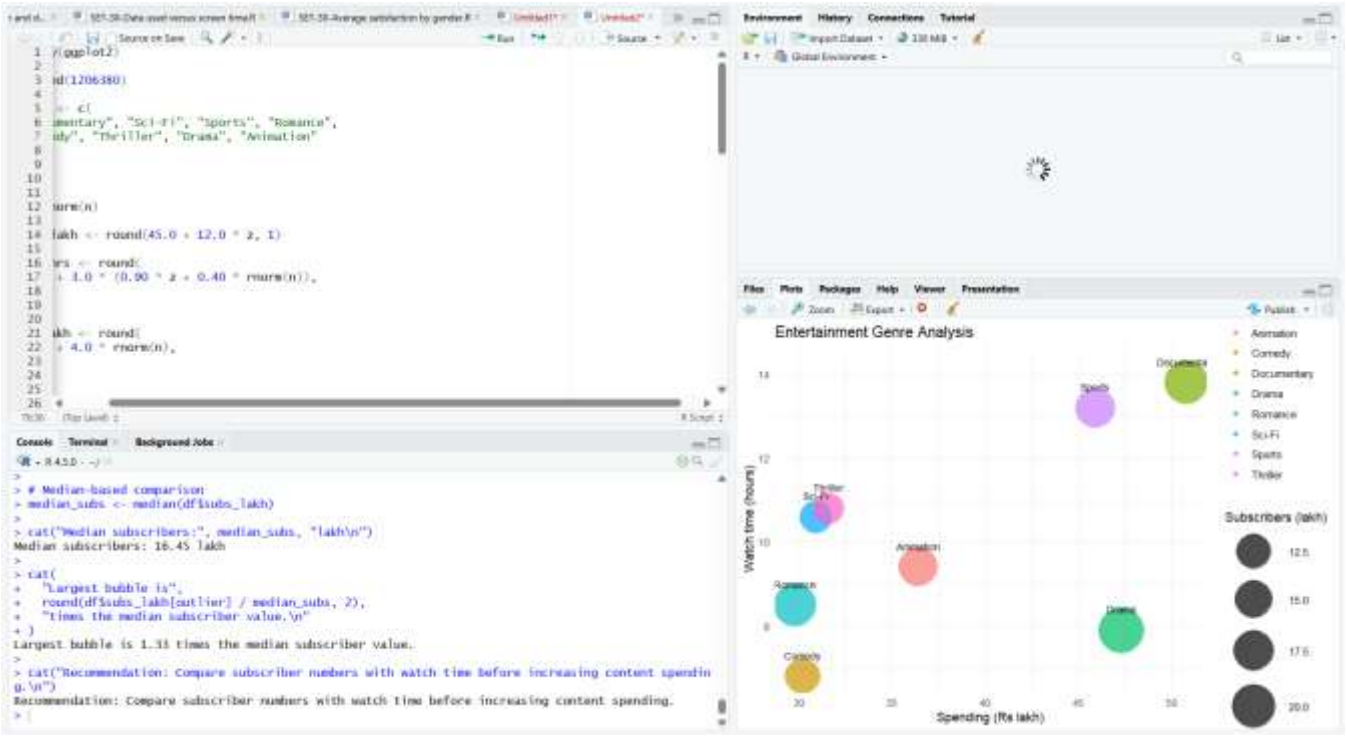
Step 1: Run the sheet's code unchanged with set.seed(1206380); the line subs_lakh[5] <- round(18.0 * 2.4, 1) = 43.2 (with spend 45.0 and watch time 11.0) plants Comedy.

Step 2: Draw ggplot(df, aes(spend_lakh, watch_hrs, size = subs_lakh)) + geom_point(shape = 21, fill = 'steelblue', alpha = 0.6) + scale_size_area(max_size = 16, name = 'Subscribers (lakh)') + geom_text_repel(aes(label = entity)).

Step 3: cor(spend_lakh, watch_hrs) gives the correlation of x and y.

Step 4: z-score of the size = (subs_lakh - mean(subs_lakh)) / sd(subs_lakh); the genre with the largest absolute z is the outlier. The ratio is subs_lakh[5] / median(subs_lakh).

Program and Output



Observation

(a) Eight labelled bubbles: spend runs from 29.8 (Romance) to 50.8 lakh (Documentary), watch time from 7.9 (Drama) to 13.8 hours (Documentary) and subscribers from 10.4 (Thriller) to 43.2 lakh (Comedy).

(b) The correlation between spend and watch time is r = +0.458, a moderate positive link: genres with more marketing spend tend to have somewhat longer watch time, though Drama (47.3, 7.9) breaks the pattern.

(c) The outlier on size is Comedy. Its position is ordinary: spend 45.0 (the mean is 39.71) and watch time 11.0 (the mean is 10.65), but it has 43.2 lakh subscribers. With a mean of 19.81 and an SD of 10.28, its z-score = (43.2 - 19.81) / 10.28 = +2.28; the other seven genres have z-scores between -0.92 and +0.20.

(d) The median subscribers of the 8 genres is 17.85 lakh, so Comedy has 43.2 / 17.85 = 2.42 times the median. A plain scatterplot of spend against watch time would draw Comedy at (45.0, 11.0), in the middle of the cloud between Sports (45.9, 13.2) and Drama (47.3, 7.9), so it would look like any other genre and its large subscriber base would never be seen.

Recommendation: Content Head: study why Comedy has 2.4 times the median subscribers (43.2 against 17.85 lakh) with only average spend and watch time, and use its content and promotion model for the genres with fewer subscribers, while working on its watch time (11.0 hours), which is below Documentary (13.8) and Sports (13.2).

Result: r(spend, watch time) = +0.458; Comedy is the outlier on subscribers (z = +2.28, 2.42 times the median) although its x/y position is near the average.

Q7: Hexbin Plot [5 marks]

Scenario: A large paired data set: loan amount (Rs lakh) (x) and applicant income (Rs thousand per month) (y) for 3000 records.

Tasks

1. Draw a hexbin plot (geom_hex(bins = 25) or the hexbin package) with a count legend and a viridis scale; also draw the plain scatterplot of the same data. [1.5]

2. Using a 10 x 10 grid (table(cut(x, 10), cut(y, 10))), find the busiest cell: give its x range, y range, count and share of all points. [1.5]

3. Report how many distinct (x, y) pairs there are (nrow(unique(df))) and explain why the scatterplot alone would mislead. [1]

4. Give one recommendation for the Zonal Manager. [1]

Aim: To show where 3000 loan-income pairs are concentrated, which a scatterplot cannot show because the points overlap.

Why this chart: With 3000 points many lie on top of each other, so a scatterplot becomes a solid blob. A hexbin plot divides the plane into hexagons and colours each by its number of points, so the density is visible, and the count legend lets the reader read how many records each colour means.

Steps

Step 1: Run the sheet's code unchanged with set.seed(1206371): about 85% of the records come from a dense cluster (loan around 38 lakh, income around 52 thousand) and the rest are spread evenly over a wide range.

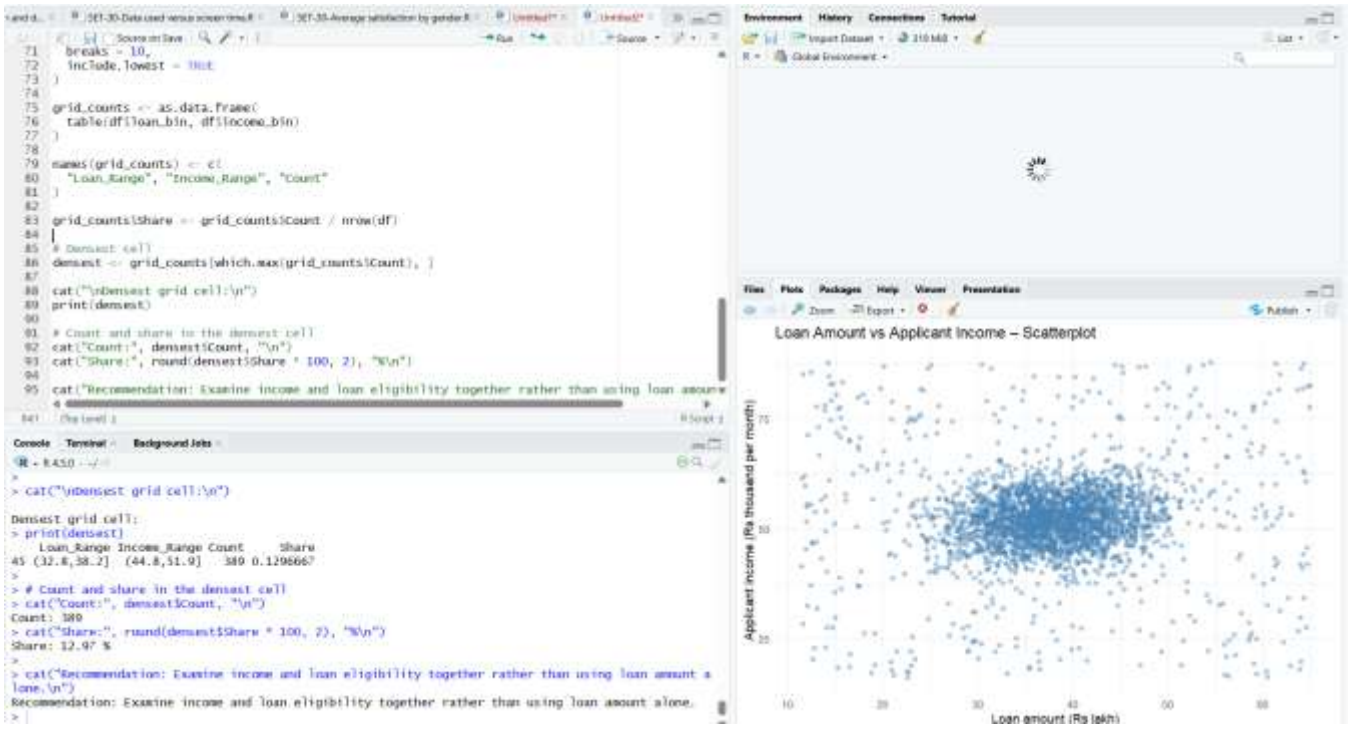
Step 2: Draw ggplot(df, aes(loan_lakh, income_k)) + geom_hex(bins = 25) + scale_fill_viridis_c(name = 'Count'), and, for comparison, ggplot(df, aes(loan_lakh, income_k)) + geom_point(alpha = 1).

Step 3: Grid count: tb <- table(cut(df$loan_lakh, 10), cut(df$income_k, 10)); the busiest cell is which(tb == max(tb), arr.ind = TRUE); its share = max(tb) / 3000 x 100.

Step 4: nrow(unique(df)) gives the number of distinct (x, y) pairs, and the number of repeated pairs is 3000 minus that.

Step 5: Compare the counts of the cells (largest, median, smallest) to see how much the scatterplot hides.

Program and Output



Observation

(a) The loan amounts run from 11.4 to 65.0 lakh (mean 37.8) and the incomes from 16.1 to 87.8 thousand (mean 52.2). The hexbin plot has 432 occupied hexagons, the densest holding 85 points, and shows one bright core on a pale background; the scatterplot shows a solid blob in the middle.

(b) Busiest 10 x 10 cell: loan between 32.8 and 38.2 lakh and income between 51.9 and 59.1 thousand, with 377 points, which is 12.57% of all 3000. The next cells have 370 (income 44.8 to 51.9), 346 and 325 points; the five busiest cells (5% of the 100 cells) hold 54.1% of all points, and 12 cells (loan 27.5 to 48.9, income 44.8 to 66.3) hold 75.2%.

(c) There are 2907 distinct (x, y) pairs out of 3000 records, so 93 records (3.1%) repeat another record's exact values and are drawn on top of each other. The scatterplot alone misleads because the cells run from 1 to 377 points (the median occupied cell has only 6, the busiest is 63 times larger) and a solid blob cannot show it; 45 of the 100 cells have 5 points or fewer but they look as important as the core, and the overall correlation (-0.05) says nothing about the dense core.

(d) In the hexbin plot the colour carries the count, so the core, the dense bands around it and the thin tail are all visible and can be read from the legend.

Recommendation: Zonal Manager: plan the standard loan product around the dense core (loan 27.5 to 48.9 lakh, income 44.8 to 66.3 thousand), which holds 75% of the records, and handle the thinly spread records outside it (45 of the 100 cells have 5 or fewer points) case by case; use hexbin plots, not scatterplots, for reports on large data.

Result: The busiest cell is loan 32.8 to 38.2 lakh and income 51.9 to 59.1 thousand with 377 points (12.57%); there are 2907 distinct pairs out of 3000, so a scatterplot would hide the density.

Summary of Answers

Question	Main answers
Q1	r = -0.647, slope -8.51; outlier row 5 (88.5, 1630; rstandard 4.28); linear (R-squared 0.419) fits better than quadratic (0.438, p = 0.41)
Q2	Strongest pairs: tower_density - churn_pct -0.82 and tower_density - traffic_tb -0.74; drop_rate - churn_pct -0.03 (no link)
Q3	PC1 31.71%, PC2 25.07%, 4 components for 80%; footfall_k largest PC1 loading (0.591); group R-squared 0.570 (PC1) and 0.008 (PC2); t-SNE no better than PCA
Q4	8 edges, density 0.381, 1 component; Pune most central (degree 5, betweenness 11); deleting Pune leaves 3 components
Q5	3 improved, 3 worsened, 1 flat; largest change Orthopaedics +7.9 (18.2%); paired t-test t = 0.163, p = 0.876, mean change +0.31
Q6	r = +0.458; Comedy outlier on subscribers, z = +2.28, 2.42 times the median
Q7	Busiest cell loan 32.8 to 38.2, income 51.9 to 59.1, 377 points (12.57%); 2907 distinct pairs of 3000

Conclusion

The seven charts each answered a different question about relationships: a scatterplot showed that one wrong record weakens an otherwise strong link (r from -0.91 to -0.65), a correlogram showed that tower density, not call drops, goes with churn, PCA showed that one component carries the group difference, a network showed that Pune holds the parcel network together, a slope chart and a paired test showed that the process change had no consistent effect, a bubble chart exposed Comedy's large subscriber base, and a hexbin plot showed where 3000 points really lie. Each statement is backed by a value from the output, so the recommendations rest on evidence.